Training Data Overview

Data Composition and Transformation Pipeline for Phase 0 (C1-C4 Framework)

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Overview

This document provides a comprehensive overview of the training data pipeline for Phase 0, documenting how raw source data (`us_labels`, `us_shocks`, `us_body`) is transformed into evaluation-ready datasets for the C1-C4 codebook framework.

Purpose:

- Transparently describe all data transformations
- Document observation-level changes (passage to act, document to chunk, etc.)
- Explain key assumptions and design decisions
- Visualize data flow and composition at each stage

Key Transformation:

The most significant transformation is from **passage-level** (us_labels with 340 passages describing 44 unique acts) to **act-level** (aligned_data with 44 rows, one per act). Multiple passages describing the same act are grouped and concatenated into a single observation.

Data Composition and Transformation Pipeline

This section documents how the raw source data (`us_labels`, `us_shocks`, `us_body`) is transformed into evaluation-ready datasets. We describe the transformation logic, observation-level changes, key assumptions, and provide visual summaries at each stage.

Data Flow Overview

The data flows through multiple transformation stages, changing observation levels and structure:

Data Transformation Pipeline Overview

Observation levels and row counts at each stage

Dataset	Unit of Observation	Row Count	Unique Acts	Purpose
us_labels	Passage (multiple per act)	376	40	Human-labeled passages describing acts
us_shocks	Quarter (multiple per act)	90	49	Romer & Romer shock magnitudes/ timing
us_body	Document (one per year-body)	360	—	Extracted text from government PDFs
aligned_data	Act (one row per act)	39	39	Joined labels + shocks at act level
chunks	Chunk (sliding window over pages)	15,061	—	Sliding-window document chunks for LLM input
c1_chunk_data	Tiered chunks (list: tier1, tier2, negatives)	13,688	39	Tier-assigned chunks for C1 LOOCV evaluation

Key Transformation:

The most significant transformation is from **passage-level** (us_labels) to **act-level** (aligned_data). Multiple passages describing the same act are grouped and concatenated into a single observation.

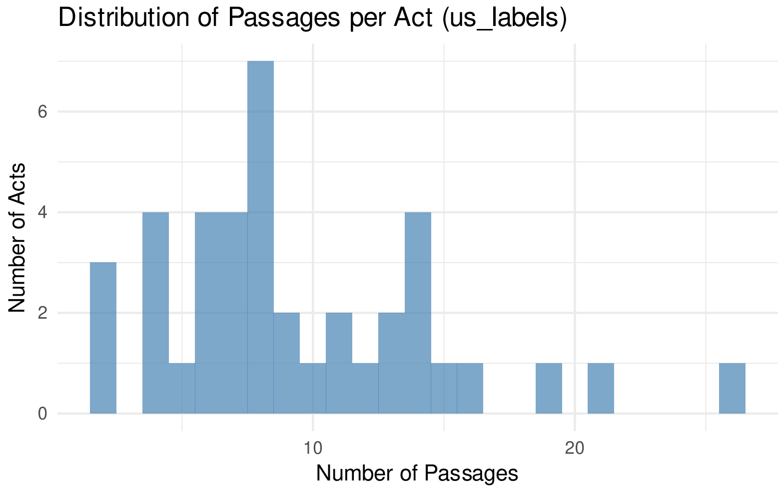
Component 1: Source Data (us_labels)

Observation Level: Passage (multiple rows per act)

Structure: Human-labeled passages from Romer & Romer (2010) dataset describing motivations for US fiscal policy acts.

us_labels: Passage-Level Structure

Metric	Value
Total passages	376
Unique acts	40
Passages per act (min/median/max)	2 / 8 / 26
Motivation categories	4
Date range	1945-01-03 to 2003-01-11



Passage Distribution by Category (us_labels)

Motivation Category	Passages	Percent
Long-run	163	43.4%
Countercyclical	88	23.4%
Spending-driven	64	17.0%
Deficit-driven	61	16.2%

Example Passages (Act with Multiple Passages):

Example: Revenue Act of 1945 (14 passages)

Passage 1:

> full employment in peacetime can be assured only when the reduction in war demand is approximately offset by additional peacetime demand from the millions of consumers, businesses, and farmers

Passage 2:

> we must overhaul the wartime tax structure to stimulate consumers' demand and to promote business investment. The elements of such a tax program should be developed now so that it can be put into effect after victory

Passage 3:

> I recommend that a transitional tax bill be enacted as soon as possible to provide limited tax reductions for the calendar year 1946. … [T]he new bill should aim principally at removing barriers to speedy reconversion and to the expansion of our peacetime economy

Key Variables:

- `act_name`: Name of the fiscal act
- `motivation`: Text passage describing the motivation
- `category`: Motivation category (Spending-driven, Countercyclical, Deficit-driven, Long-run)
- `exogeneity`: Exogenous or Endogenous classification
- `date`: Date of the act
- `source`: Source document for the passage

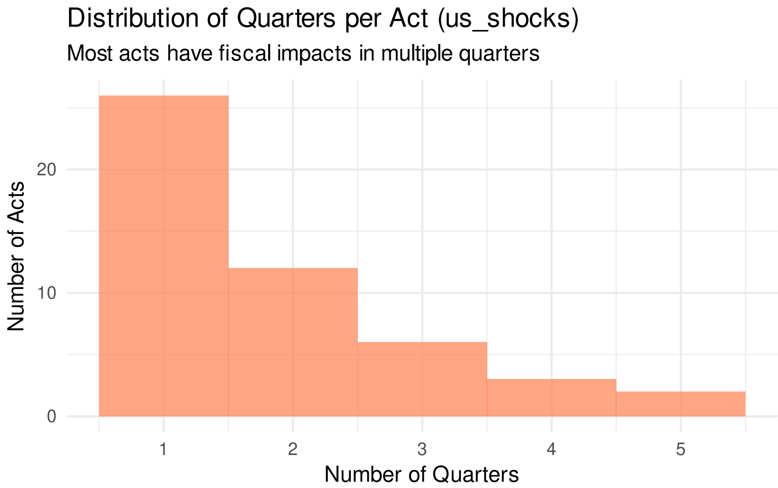
Component 2: Source Data (us_shocks)

Observation Level: Quarter (multiple rows per act, one per quarter of fiscal impact)

Structure: Romer & Romer (2010) dataset with magnitudes and timing of fiscal shocks.

us_shocks: Quarter-Level Structure

Metric	Value
Total quarters/shocks	90
Unique acts	49
Quarters per act (min/median/max)	1 / 1 / 5
Date range	1945-11-08 to 2003-05-28



Example: Act with Multiple Quarters

Example: Social Security Amendments of 1950

Multiple quarters of fiscal impact

Act	Date Signed	Impact Quarter	Magnitude	Category
Social Security Amendments of 1950	1950-08-28	1951-01-01	$0.30B	Spending-driven
Social Security Amendments of 1950	1950-08-28	1954-01-01	$1.30B	Deficit-driven
Expiration of Excess Profits Tax and of Temporary Income Tax Increases	1954-01-01	1954-01-01	$−3.70B	Spending-driven
Expiration of Excess Profits Tax and of Temporary In-	1954-01-01	1954-01-01	$−1.30B	Deficit-driven

Example: Social Security Amendments of 1950

Multiple quarters of fiscal impact

Act	Date Signed	Impact Quarter	Magnitude	Category
come Tax Increases				
Social Security Amendments of 1958	1958-08-28	1959-01-01	$1.10B	Spending-driven

Key Variables:

- `act_name`: Name of the fiscal act
- `date_signed`: Date the act was signed
- `change_in_liabilities_quarter`: Quarter of fiscal impact
- `change_in_liabilities_billion`: Magnitude in billions of dollars
- `present_value_billion`: Present value of impact
- `change_in_liabilities_category`: Motivation category
- `change_in_liabilities_exo`: Exogenous flag

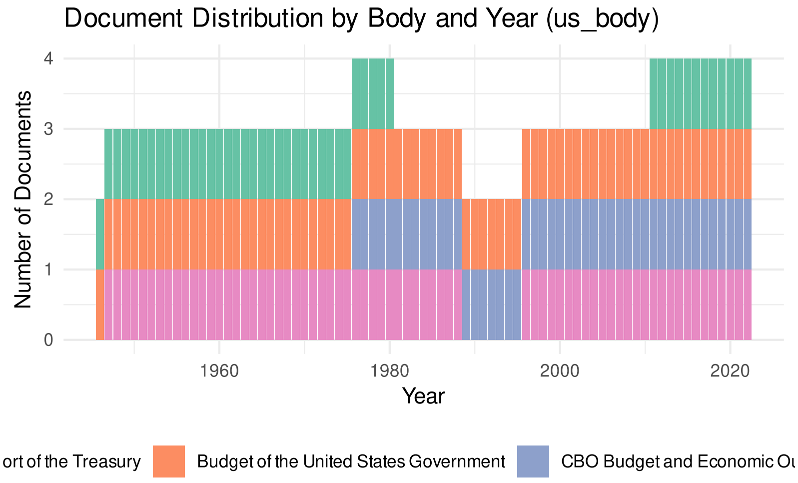
Component 3: Source Data (us_body)

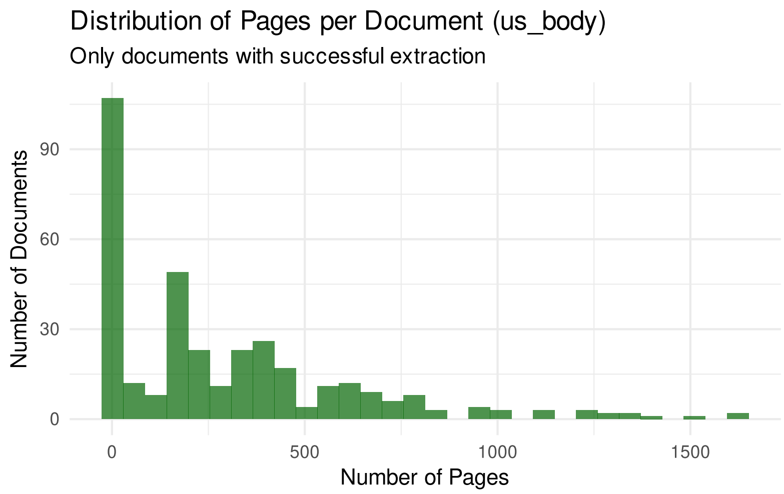
Observation Level: PDF volume (one row per extracted PDF file)

Structure: Extracted text from US government PDF documents (Economic Reports, Budget documents, Treasury Reports). Each row is one PDF file. Most documents correspond to a single PDF, but multi-volume Budget documents (2006-2009) produce multiple rows sharing the same `package_id`. The `package_id` is the unique document identifier carried forward into the chunking pipeline.

us_body: PDF Volume Structure

Metric	Value
Rows (PDF volumes)	360
Unique documents (package_id)	246
Rows with extracted text (n_pages > 0)	350 (97.2%)
Extraction success rate	97.2%
Year range	1946 to 2022
Document bodies	4





Purpose: Source corpus for chunking. Chunks serve as LLM inputs for C1 evaluation and production inference.

Component 4: aligned_data (Labels + Shocks)

Transformation: Passage-level to Act-level

Process: Join `us_labels` with `us_shocks` to combine human motivations with shock magnitudes/timing.

Key Transformation Logic (from `align_labels_shocks()` in `R/prepare_training_data.R`):

1. **Act name cleaning:** Normalize whitespace in both datasets
2. **Exact matching:** Join on cleaned act names
3. **Fuzzy matching:** For unmatched acts, use Jaro-Winkler similarity (threshold = 0.85)
4. **Collapse quarters:** Use first quarter's data as representative (us_shocks has multiple rows per act)
5. **Group passages:** Concatenate all passages for each act with `\n\n` separator

Assumptions:

- **One representative quarter:** When acts have multiple quarters of impact (us_shocks), we use the first quarter's magnitude and timing as the "primary" shock. This is an acceptable simplification because the LOOCV evaluation tests measure identification, not multi-quarter extraction.
- **Passage concatenation:** All passages describing the same act are concatenated into a single text field. This preserves all human-labeled context while creating a single observation per act.
- **Fuzzy matching threshold (0.85):** Acts with name similarity of at least 0.85 are considered matches. This accommodates minor spelling variations (e.g., "Revenue Act of 1964" vs "Revenue Act, 1964").

Transformation: Passage/Quarter to Act Level

Observation level change in aligned_data

Dataset	Unit	Rows	Acts
us_labels (passages)	Passage	376	40
us_shocks (quarters)	Quarter	90	49
aligned_data (acts)	Act	39	39

Example: Concatenated Passages per Act

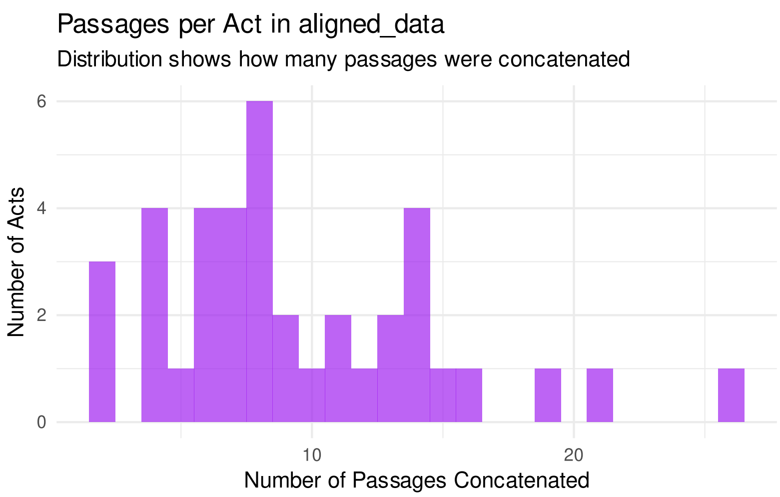
Multiple passages combined into single text field

Act Name	# Passages	Chars	Text Preview
Changes in Depreciation Guidelines and Revenue Act of 1962	7	1169	When I came into office in January 1961 this country was in a recession. We have made a recovery from that recession And now we must be concerned ...
Crude Oil Windfall Profit Tax Act of 1980	9	1314	In order to prevent oil producers from reaping excessive profits from decontrol a windfall profits tax is proposed A windfall profits tax is the o...
Deficit Reduction Act of 1984	14	1908	A major reduction in the structural budget deficit must therefore be achieved over the

Example: Concatenated Passages per Act

Multiple passages combined into single text field

Act Name	# Passages	Chars	Text Preview
			next several years to develop a 'down payment' package that...
Economic Growth and Tax Relief Reconciliation Act of 2001	6	1540	the surplus is the people's money. And our people are overtaxed. ... [W]ith about quarter of that surplus, we need [to] give the hard-working people ...
Economic Recovery Tax Act of 1981	19	2936	I have long advocated a 30 percent reduction in income tax rates [w]e are taxing ourselves into economic exhaustion and stagnation, crushing our a...



Example: Before and After Alignment

Before Alignment: Changes in Depreciation Guidelines and Revenue Act of 1962
Source: us_labels (passage-level, 7 rows)

Passage 1:

> When I came into office in January 1961 this country was in a recession. We have made a recovery from that recession

Passage 2:

> And now we must be concerned with the forward movement of our economy. The level of our economy … is high but, considering all the resources which this country has, it should be higher

Passage 3:

> measures which I think would speed up our economy, which are designed to give us more jobs and more growth

Passage 4:

> will stimulate investment in capacity expansion and modernization, contribute to the growth of our productivity and output, and increase the competitiveness of American exports in world markets

Passage 5:

> help to stimulate the investment needed for sustained expansion and longer-run growth

Passage 6:

> the investment credit … is designed to provide a stimulant to the economic growth of this country. This is needed both to improve our competitive position abroad and in the long run to raise our standard of living at home. On the other hand, the other provisions of this bill are designed to impro…

Passage 7:

> the investment credit, coupled with the liberalized depreciation, will provide a strong and lasting stimulus to a high rate of economic growth

After Alignment

Source: aligned_data (act-level, 1 row)

- **Passages concatenated:** 7
- **Total characters:** 1,169
- **Motivation category:** Long-run
- **Magnitude:** $−1.35B
- **Exogenous:** TRUE

Concatenated text (first 500 chars):

> When I came into office in January 1961 this country was in a recession.
> We have made a recovery from that recession

And now we must be concerned with the forward movement of our economy. The level of our economy … is high but, considering all the resources which this country has, it should be higher

measures which I think would speed up our economy, which are designed to give us more jobs and more growth

will stimulate investment in capacity expansion and modernization, contribute to the ……

Alignment Quality:

Alignment Statistics

Metric	Value
Acts in us_labels	40
Acts in us_shocks	49
Acts successfully aligned	39
Alignment rate	97.5%

Component 5: chunks (Sliding Window Document Chunks)

Transformation: Document-level to Chunk-level (sliding window)

Process: Split each document in `us_body` into overlapping chunks that fit within the LLM context window, using `make_chunks()` in `R/make_chunks.R`.

Key Logic:

1. **Sliding window:** 10-page window with 3-page overlap between consecutive chunks
2. **Token limit:** 40,000 tokens (advisory; warns but does not split)
3. **Page separators:** Pages joined with `--- PAGE BREAK ---` markers for page number tracking
4. **Token estimation:** Characters / 4 as approximate token count

Design Rationale:

Chunks are the unit of input for C1 evaluation and production. Chunk-based evaluation tests the model under production conditions: long documents containing fiscal measures amid economic commentary, forecasts, and administrative content. This matches the H&K "eval-must-match-production" principle.

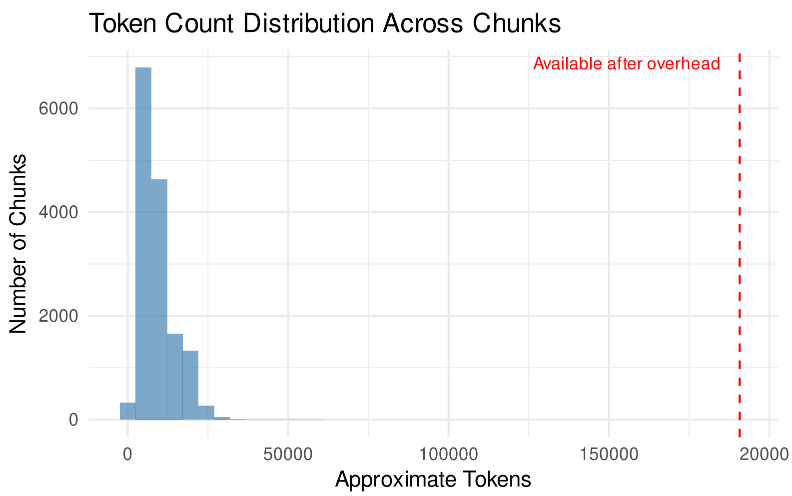
Context window budget:

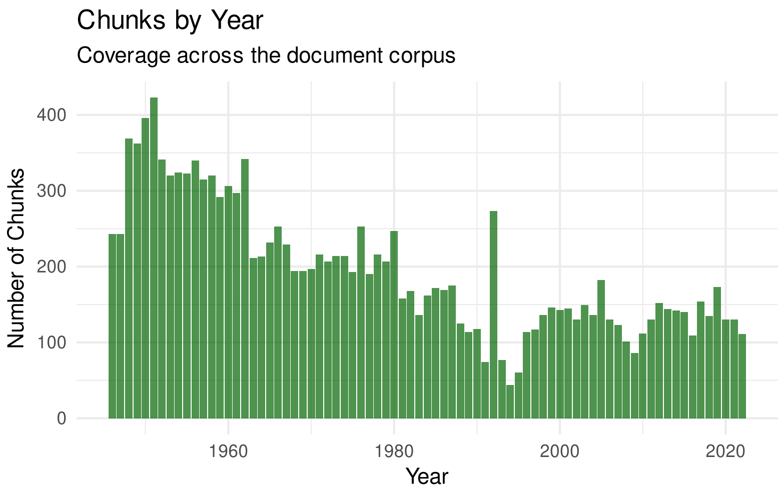
With 10-page windows, chunks are well within the 40K token advisory limit. With prompt overhead (~9K tokens for codebook, few-shot examples, and system instructions) and a 2K output buffer, all chunks fit comfortably within the 200K context window used by both Haiku 3.5 (S1/S2/S3 evaluation) and Sonnet/Opus (production).

Component	Approx. tokens
Codebook (YAML)	4,100
Few-shot examples (~5)	3,100
System prompt	125
Output buffer	2,000
Overhead total	**9,300**
Available for chunk	**190,700**
Largest chunk in corpus	**155,277**

chunks: Sliding Window Structure

Metric	Value
Total chunks	15,061
Source PDF volumes (rows in us_body)	360 (⟶ all 360 chunked)
Unique documents (package_id)	245 (matches us_body above)
Chunks per document (min/median/max)	4 / 50 / 250
Token count (min/median/max)	26 / 7,599 / 61,058
Pages per chunk (min/median/max)	1 / 10 / 10





Component 6: c1_chunk_data (Positive and Negative Chunks for C1 Evaluation)

Transformation: Identify which chunks are positive (contain a known fiscal measure) and which are negative (contain no fiscal policy discussion), for use in C1 LOOCV evaluation.

Process: `prepare_c1_chunk_data()` in `R/identify_chunk_tiers.R` classifies each chunk based on its relationship to the 44 labeled acts in `aligned_data`.

Positive identification (two tiers):

We identify positive chunks at two levels of stringency:

1. **Tier 1 (verbatim passage):** Search for the first 100 characters of each labeled passage (from `aligned_data$passages_text`) as a substring within year-adjacent chunks. Fallback to 60 characters if no match. These are gold-standard positives: we know the exact Romer and Romer (2010) passage text appears in the chunk.
2. **Tier 2 (act name mention):** Search remaining chunks for act name matches using three strategies: whitespace-normalized substring matching (handles OCR line breaks), subcomponent decomposition for compound act names (split on "and", parenthetical extraction, embedded formal names, Public Law numbers), and co-occurrence matching for event-type names (e.g., "expiration" AND "excess profits"). All strategies require at least 2 fiscal keyword co-occurrences to filter incidental mentions.

Tier 2 adopts a deliberately lax inclusion criterion. Relying on Tier 1 alone would limit positive chunks to those containing verbatim passages from Romer and Romer (2010), which are short excerpts that the original authors selected to illustrate each act's motivation. Many chunks that substantively discuss an act's magnitude, timing, or implementation would be missed. By including Tier 2, we ensure the model sees the full range of fiscal discussion surrounding each act, which is essential for downstream codebooks (C2 motivation, C3 timing, C4 magnitude) that require richer context than the labeled passages alone provide.

Negative identification:

3. **Negative:** All chunks not assigned to Tier 1 or Tier 2. Each negative is tagged with a continuous `key_density` score (fiscal keyword count / total words) measuring fiscal vocabulary concentration. This replaces the previous binary relevance-key exclusion with a gradient suitable for stratified error analysis in S3.

Evaluation targets:

- **Tier 1 Recall** (target: 95%) measures whether the model can find known fiscal measures embedded in long documents. Failures here indicate a fundamental problem.
- **Tier 2 Recall** measures generalization beyond verbatim labeled text. Lower recall is expected and informative.
- **Combined Recall** (target: 90%) is the primary gating metric for the C1 funnel.
- **Negative chunks** test precision: can the model correctly reject chunks that do not describe a specific fiscal measure? Negatives span a gradient from clean (zero fiscal vocabulary) to fiscally rich (high key density), and `key_density` enables stratified precision analysis in S3 to identify the threshold where the model's discrimination breaks down.

C1 Chunk Tier Distribution

How chunks are assigned to evaluation tiers

Tier	Chunks	Percent
Tier 1	70	0.5%
Tier 2	1986	15.1%
Negative	11632	88.7%
Total	13112	100.0%

Tier 1: Verbatim Passage Matches

Metric	Value
Tier 1 chunks (total)	70
Acts with Tier 1 matches	30
Acts without Tier 1 matches	9
Chunks per act (min/median/max)	1 / 2 / 7
Avg tokens per Tier 1 chunk	6,873

Tier 2: Act Name Mention Matches

Metric	Value
Tier 2 chunks (total)	1986

Tier 2: Act Name Mention Matches

Metric	Value
Acts with Tier 2 matches	39
Chunks per act (min/median/max)	4 / 44 / 114
Avg tokens per Tier 2 chunk	7,776

Negative Chunks by Source Type

Stratified by document body

Source	Chunks	Percent
Budget	5696	49.0%
Treasury	3054	26.3%
ERP	2399	20.6%
Other	483	4.2%

Table 1: Negative Chunk Key Density Distribution

Key Density Bin	Chunks	Percent
0	517	4.4%
(0, 1%]	6129	52.7%
(1%, 3%]	4529	38.9%
(3%, 5%]	407	3.5%
>5%	50	0.4%

Verification: Are Negatives Correctly Excluded from Positives?

Negatives now intentionally include chunks with fiscal vocabulary (measured by `key_density`). The verification question is not "are negatives clean?" but "are negatives correctly excluded from positives?" We check whether any negative chunks contain a known act name that should have been caught by Tier 2. A small number is acceptable: act names appearing outside the ±5 year window or without 2+ fiscal keywords are correctly excluded from Tier 2.

Negative Chunk Contamination Check

Verify no act names were missed by Tier 2 matching

Check	Result	Status
Negative chunks sampled	100	—
Contains known act name	0 (0.0%)	Pass
Contains 'enacted/signed into law'	34 (34.0%)	Review

> **Note**
>
> Negatives are all chunks not assigned to Tier 1 or Tier 2. Each negative is tagged with `key_density` (fiscal keyword concentration) for downstream stratification. Some negatives contain fiscal vocabulary by design; the key question is whether any contain a known act name that Tier 2 should have caught. Act name mentions in negatives are expected when they fall outside the ±5 year window or lack the required 2+ fiscal keyword co-occurrences.

Summary: Key Transformations

Data Transformation Summary

Key changes in observation level and row counts

Transformation	Level Change	Operation	Input Rows	Output Rows
us_labels → aligned_data	Passage-level → Act-level	Group passages by act, concatenate	376	39
us_shocks → aligned_data	Quarter-level → Act-level	Use first quarter as representative	90	39
us_body → chunks	Document-level → Chunk-level	Sliding window (10 pages, 3 overlap, 40K token limit)	360 docs	15,061
chunks → c1_chunk_data	Chunk-level → Tiered evaluation sets	Verbatim match (T1), name/sub-component/co-occurrence match (T2), all remaining (Neg)	15,061	T1: 70, T2: 1986, Neg: 11632

Critical Findings

Finding 1: Dataset Size

44 labeled acts are available in `us_labels`, not 126 from the full `us_shocks` dataset. This is the number of acts with human-labeled motivation passages, which is the ground truth for all codebook evaluation.

Impact on C1-C4:

- **C1 (Measure ID):** LOOCV on 44 acts with chunk-based inputs. Targets: Combined Recall at least 90%, Tier 1 Recall at least 95%, Precision at least 70%.
- **C2 (Motivation):** With only 6 Countercyclical acts, this category will have limited evaluation power. LOOCV helps by using all 44 acts.
- **C3/C4 (Timing/Magnitude):** 41 of 44 acts have complete timing and magnitude data.
- **Few-shot budget:** Each LOOCV fold uses 43 acts for training. With 5 examples per class, this consumes only ~12% of available acts.

Finding 2: Passage Concatenation Creates Richer Context

- **Median 8 passages per act** (range 1-25), yielding roughly 5K tokens per act
- More context improves motivation classification (C2) and information extraction (C3/C4)
- Claude's 200K context window easily handles the longest acts
- Higher API costs per call, but LOOCV keeps total calls manageable (44 folds)

Finding 3: Multi-Quarter Acts

- us_shocks shows **median 7 quarters per act**, max 32 quarters
- aligned_data uses only the **first quarter** as representative
- C3 (Timing) and C4 (Magnitude) will need to handle multi-quarter extraction in their codebook design
- This is a known simplification; the C3/C4 output schemas should support multiple implementation phases

Finding 4: Chunk Tier Coverage

C1 Chunk Tier Coverage

How many acts have chunks assigned to each tier

Metric	Value
Acts with Tier 1 (verbatim) coverage	30 / 39 (77%)
Acts with Tier 2 (name mention) coverage	39 / 39 (100%)
Acts with any chunk coverage	39 / 39 (100%)
Total labeled acts	39

Tier 1 coverage indicates how many labeled acts have their verbatim passage text recoverable in the chunked corpus. Acts without Tier 1 coverage may have passages that span chunk boundaries or come from documents not yet extracted. Tier 2 provides additional coverage through whitespace-normalized substring matching, subcomponent decomposition for compound act names, and co-occurrence matching for event-type names. See `notebooks/verify_chunk_tiers.qmd` for detailed per-act analysis of matching mechanisms and temporal consistency.

Conclusion

The training data is **ready for Phase 0 codebook evaluation** with the C1-C4 framework.

Strengths:

- 100% alignment rate between labels and shocks
- Chunk-based evaluation matches production conditions (long documents, fiscal measures amid noise)
- Three-tier system provides diagnostic granularity (Tier 1 gold standard, Tier 2 generalization, Negatives for precision)
- Tier 2 matching uses whitespace normalization, subcomponent decomposition, and co-occurrence matching to achieve near-complete act coverage
- Negative chunks tagged with continuous `key_density` for stratified precision analysis in S3 (no arbitrary relevance-key exclusion)
- Rich passage context (median 8 passages per act) preserved in aligned_data for few-shot examples
- LOOCV design uses all 44 acts for both training and evaluation

Known Limitations:

- **44 acts total:** Small sample, but LOOCV maximizes use of available data
- **Countercyclical under-representation:** Only 6 acts in this motivation category
- **First-quarter simplification:** Multi-quarter acts reduced to single representative
- **Tier 1 coverage gaps:** Some acts may lack verbatim passage matches if passages span chunk boundaries or come from unextracted documents
- **No ground truth for Malaysia:** Phase 2 requires expert validation (target: at least 80% agreement on C1)

Evaluation Strategy: Each codebook (C1-C4) is evaluated through the Halterman & Keith 5-stage framework: S0 (codebook prep), S1 (behavioral tests), S2 (LOOCV on 44 acts with bootstrap CIs), S3 (error analysis and ablation), with S4 (fine-tuning) reserved as a last resort. C1 uses chunk-level inputs for S1-S3, with passage-level few-shot examples and chunk-level test inputs.

Romer, Christina D., and David H. Romer. 2010. "The Macroeconomic Effects of Tax Changes: Estimates Based on a New Measure of Fiscal Shocks." *American Economic Review* 100 (3): 763–801.